MedZen Speech Platform

Project Handoff Report — Claude (sole implementation engineer + reviewer)

Date: 2026-08-20 **Repo:** ~/Documents/medzen-platform (github.com/fotso94/medzen-platform, private) **HEAD:** 6d7d68c

AWS account: 558069890522 **Region:** eu-central-1 (all resources)

Purpose: allow another AI assistant or senior engineer to independently understand, audit and continue this project without prior conversations. Written for a follow-up independent audit by ChatGPT.

Method: every load-bearing claim below was re-verified live on 2026-08-20 against git, the full test suite, S3, ECR, EC2, SageMaker, IAM, SSM, Secrets Manager, EKS and Cost Explorer — not reproduced from memory. Claims that could not be re-verified are explicitly marked NOT VALIDATED. This verification pass itself found and fixed one latent deploy-day bug (§11, item 8).

Read this first: §19 HANDOFF — START HERE (last page) is the entry point for the next engineer. §3 is the one-page status dashboard.

1. Project objective and original architecture

MedZen is building a **medical-assistant speech platform for African languages**: a patient speaks in their language; the platform transcribes (ASR), retrieves grounded medical guidance (RAG), generates a cited answer (LLM), and speaks it back (TTS). Target: ~20 in-scope languages for generation, 40+ languages with frozen evaluation sets.

The plan is phased: **B1** contracts & schemas → **B2** data ingest (A3 manifest standard) → **B3** per-language baselines → **B4** frozen evaluation suite → **B5** training (fine-tuned ASR per language) → **B6** serving (5-service platform on EKS) → **B7** CI/CD activation. Base ASR model: Meta omniASR (CTC 1B family). LLM: AWS Bedrock. TTS: Fish Audio (bought voices) until in-house TTS data exists.

Current architecture (all five service images built and pushed; not yet deployed): client → **speech-orchestrator** (API-key auth via Secrets Manager, SSM registry routing) → **asr-runtime** (omniASR + approved fine-tuned weights from S3) → **rag-index** → **llm-gateway** (Bedrock Claude Haiku 4.5, citation-bound JSON contract) → **speech-tts-gateway** (Fish Audio, per-language voice registry in SSM). Every gateway defaults to an offline fake mode; real modes are explicit env opt-ins; unknown modes refuse (fail-closed).

2. Governance: who built what, and when Claude took control

Until 2026-08-14 the project was implemented by **ChatGPT/Codex** with **Claude as independent reviewer** (per-service exit reviews, PASS/CONCERNS/HOLD verdicts). On **2026-08-15** the owner transferred the implementation role: Claude became **sole engineer + self-reviewer**.

> **Evidence:** ~/Documents/medzen-shared/codex_status.txt line 1: “Report ID: CLAUDE-ENGINEER-STATUS-2026-08-15-001 … Engineer: Claude (sole engineer + self-reviewer per owner role transition 2026-08-15)”. Git: 756 commits dated ≤2026-08-14, 320 commits since 2026-08-15 (total 1,074+ since 2026-07-28).

Standing owner constraints that bind any successor:

• Signed data agreements (LIC-2026-002 ShareAlike, LIC-2026-003 africanvoices, LIC-2026-004 NaijaVoices) are **PRIVATE with the owner** — repo carries attestation records only; never commit or upload the papers.
• The live ECS service **medzen-tts-dev** (cluster medzen-tts-dev, service medzen-tts-gateway) is the working reference/fallback — **never modify it**; review only.
• The engineering agent never handles credentials (HF login and API keys are loaded by the owner directly).
• Cost reporting: on-demand pricing as basis; add $100 to every reported cost maximum.
• Training spend requires an owner-approved packet: bindings JSON → render → validate → written authorization phrase in the shared file → launch. Job names are single-use.

3. Current phase and overall status (2026-08-20, ~03:45 UTC)

Area	Status	Evidence (verified 2026-08-20)
Test suite	2,202 passed, 1 skipped, 7 deselected (92 s)	pytest run 2026-08-20 on HEAD
Shipped model	kinyarwanda ASR v1 APPROVED: 0.372 WER / 0.089 CER vs 0.410 / 0.097 zero-shot baseline (9.3% rel. gain)	s3://medzen-speech/…/kinyarwanda-v1/model.pt (1.95 GB)
Serving images	All 5 platform images in ECR, provenance-stamped, container … EKS nodegroups at 0/4, ek…	ECR digests §13; … NOT deployed …
In flight 1	kinyarwanda 2,000 h re-pull: decode DONE, uploading 378,700 / 1,018,628 clips (~37%)	box i-0…; …/research/b5-training/rw-ingest/host.lo…
In flight 2	Pidgin (africanvoices): 7/42 archives in S3 (owner Mac uploading … left); self-sequencing ingest box running	box i-03cd7a4…; Mac … log "STAGING Batch_1"
In flight 3	NaijaVoices: hausa COMPLETE (113,439 clips); igbo uploading 35,000/92,083; yoruba queued	curated/hausa/ in S3; Mac ingest log
In flight 4	SageMaker quota 1→2 concurrent g6.xlarge training jobs: CASE_OPENED (AWS Demand Review)	request …; L-56AE9D73
Next gate	kinyarwanda v2 FULL fine-tune on ~2,000 h once re-pull lands; ship as v2 only if relative gain ≥ 9.3%	owner decision 2026-08-19 (verbatim in kinyarwanda-v1.yaml)

Phase: B5 (training) mid-campaign + B6 (serving) built-and-proven but dark. Nothing about the serving stack is live; the only running cost sources are the two ingest boxes and the EKS control plane (§15).

4. What existed before Claude took control (verified, not assumed)

Built in the Codex era (2026-07-28 → 2026-08-14, 756 commits), verified present and working today:

• **A3 manifest standard + ingest pipeline core** (pipeline/ingest.py, pipeline/adapters/base.py): checksum-verified rows, licence tiers, speaker/text-disjoint splits, byte-dup refusal, near-dup validator.
• **Adapters**: soreva (frozen eval sets), waxalnlp, aaf, lrsc, yemba_egra, meta_omnilingual, kallaama, common_voice (base).
• **Frozen evaluation pools** under s3://medzen-speech/eval/ for 40+ languages (SOREVA et al.) — these are the benchmark; training data must never leak into them.
• **Baseline machinery** (scripts/run_baseline.py, paired decode comparison) and language registry (registry/languages/*.yaml, 20 generation languages).
• **B6a runtime images**: medzen-asr-runtime:b6a-003c-89f94d3, medzen-model-loader:b6a-003b-69bdce1 (ECR, 2026-08-04).
• **asr-eval-runtime image** + K8s Job eval harness on EKS medzen-speech — eval attempts 1–25 (attempt 25 failed; root cause was found by Claude, §11).
• **Contracts**: platform/contracts/speech-v1.yaml (adopted), llm-v1, tts-v1 (local-mock era), services.yaml, fixtures + JSON schemas.
• Original trainer image medzen-trainer (2026-08-01, superseded).

Evidence: git log --until=2026-08-14; ls pipeline/adapters; aws ecr describe-images (digests §13); aws s3 ls s3://medzen-speech/eval/ (40+ language prefixes).

5. What Claude implemented after taking control (320 commits, chronological)

When	Arc	What was delivered
08-15	Eval unblocked	Attempt-25 root cause (asset-card absolute /models paths not mounted) + fix + render-time mount validator; attempts 26…
08-16→17	Full evaluation suite	10 frozen shards, attempts 29–44, one attempt per authorization revision; complete zero-shot baseline pool for the 40+ e…
08-17	Training stack	pipeline/omniasr_train.py (LoRA, CTC scope, env-dict config, refusal on unknown modes) + scripts/b5_sagemaker_job.p…
08-17→18	Wave-1 + T6 + forensics	gb3 multilingual shared-adapter training (r2) → T6 gate on frozen pools: 0/9 pass, 7/9 REGRESSED → full forensics: lr 1…
08-18→19	Kinyarwanda arc	Per-language probes (clean U-curve 0.375@1800 steps) → decisive 300 h rank-32 run (dev plateau 0.355) → T6-style fu…
08-18→19	Data campaign	Tier A ingest (10 corpora: CV hausa/igbo/yoruba/english-capped, FLEURS train, WaxalNLP malagasy, NCHLT sepedi + …
08-19	Full fine-tune prep	MEDZEN_TRAIN_MODE=full in trainer (full-checkpoint save/load, all-params grad); ft image medzen-trainer-omniasr:ft-2…
08-19→20	B6 services real	llm-gateway: BedrockProvider (Converse API, eu.anthropic.claude-haiku-4-5 inference profile, JSON contract, cited-ids g…
08-20	Pidgin prep + this handoff	africanvoices adapter (XLSX quirk, fail-closed label checks) + self-sequencing ingest box; SageMaker quota request 1→…

Evidence: git log --since=2026-08-15 (320 commits); decision/evidence records in platform/decisions (270 files) and platform/evidence (379 files); codex_status.txt entries 2026-08-15 → 2026-08-20.

6. What Claude modified or corrected from previous work

Previous state (Codex era)	Correction + why
Eval image asset cards resolved checkpoints at absolute /models paths; pilot pods mounted /app only → all pilot... failed in <2 min	subPath /models mounted + render-time validator... any workload whose card
Whisper decode capped by default max_new_tokens → truncated long clips	explicit max_new_tokens; image rebuilt + republished
5 pre-existing red tests (A3 drift, registry, F placeholder, time-bomb dates)	fixed at root cause, not skipped
SSM aggregate readback exceeded parameter size on big shards	chunked readback + consumer audit
Shared multilingual adapter as the B5 training shape	abandoned on T6 evidence (0/9, interference is structural); replaced by per-language st
tts-gateway validated audio against a synthetic constant media type (would have rejected ALL real Fish audio)	media_type now derived from the provider (own deep-review catch before any real call w
Voice registry SSM path /medzen/tts/voices sat OUTSIDE the pod role's grant	moved to /medzen/registry/tts/voices (inside existing grant)
Fish key secret name medzen/fish-api-key (does not exist) baked into gateway default	found by this report's verification + image -20 default → medzen/speech/fish-api-key, runbook corre

7. What has been tested and independently verified

Claim	Verification evidence
Repo is healthy at HEAD 6d7d68c	full suite 2,202 passed / 1 skipped / 7 deselected, 92 s, 2026-08-20
kinyarwanda v1 beats baseline on the FROZEN pool	gate run on ephemeral g6.xlarge eval box, pinned image medzen-asr-eval-runtime@sha256:dafc4b14…
v1 artifact is the gated artifact	approved/kinyarwanda/asr/v1/model.pt (1,951,673,613 bytes, 2026-08-19) = merged rank-32 step-6000
All 5 serving images exist, provenance-stamped	ECR digests listed §13, SOURCE_COMMIT build-args enforced by Dockerfiles; verified via describe-ima
Orchestrator container honours the live contract	b6_live_contract_probe.py --full inside the pushed image: PASS (file + streaming), fixtures/sibling packag
Bedrock provider maps the contract	unit contract tests (stub client) + live Converse smoke during build (eu inference profile requirement disc
Tier A corpora are sound	TIER-A-REVIEW-2026-001: checksums re-verified from S3, licence labels vs sources, zero byte-leaks vs
NaijaVoices hausa corpus complete	ingest EXIT 0 2026-08-19T23:50:46Z, 113,439 clips ≈100 h, manifest in curated/hausa/asr/nv_hausa/v1/
africanvoices adapter reads the real archives	unit tests (2) + dry run against real Batch_1 rows on the Mac: 5 records pass all A3 checks

8. What exists but has NOT been fully validated

- **Serving deploy**: no image has ever run on the EKS cluster as a service (node groups at 0). Container proof ≠ cluster proof: IRSA/roles, ALB, registry namespace, secrets wiring are untested end-to-end.
- **Fish real synthesis via the platform gateway**: the real provider code is reviewed + unit-tested; no end-to-end synthesis has run because the platform Fish secret is EMPTY (owner action, §16).
- **Full fine-tune mode on real GPU at scale**: unit-tested and image-built (ft-2026-08-19a); never yet run as a SageMaker job. The v2 job will be its first live use — watch checkpoint sizes and save cadence.
- **orchestrator deployed_http_ssm mode** against real cluster services (only fixture mode has run end-to-end).
- **In-flight corpora** (rw v2 2,000 h, NV igbo/yoruba, Pidgin): not yet curated/reviewed — do not train on them until the Tier-B review and gb5 version binding exist.
- **LLM answer quality per language**: gateway contract is enforced, but the ~$5 multilingual calibration run (task #36) hasn't happened.
- **akan/serer baselines** are known-wrong (label-mapping mismatch, task #35) — do not gate anything on them.

9. Remaining tasks in recommended execution order

#	Task	Trigger / dependency
1	rw 2,000 h upload completes → curate v2 (drop_both stats, byte-leak strip vs cv17-test + other rows) → gb5 version bind	watcher on box i-04a57c0f, hours away
2	v2 FULL fine-tune packet (ft image, MEDZEN_TRAIN_MODE=full, lr ~1e-5, ~40k steps, checkpoint every 2000)	after 1; ship as v2 ONLY if eval gain > 0.3%; owner numbers approval → launch
3	Pidgin ingest completes → Tier-B deep review (NV + AV: checksums, licences, leak checks vs SCORE pidgin eval)	box i-03c84... self-fires, ~2 days
4	Next-wave decision table to owner: pidgin/hausa/igbo/yoruba proposed bars vs baselines (34%/24%/44%/51% WER) + costs	after 3; owner approval gates spend
5	akan/serer eval-baseline label fix (#35)	independent, small
6	LLM multilingual calibration ~$5 (#36)	independent, small
7	Serving deploy per B6-DEPLOY-RUNBOOK (#37): owner operating-shape decision (always-on ≈ $1,200–1,400/mo max incl. +$100 rule / business-h	owner decision required first
8	Wave-2 re-scope under per-language strategy (13 approved languages superseded in shape)	after v2 verdict

10. Training / evaluation experiments — results and conclusions

Experiment	Result	Conclusion
T5 calibration (yemba, CTC head)	PASS, <$10	trainer + packet flow proven
gb3 wave-1 shared adapter (9 langs, r2)	T6: 0/9 pass, 7/9 regressed vs zero-shot	NO_PROMOTION (B5-T6-GATE-2026-001)
Weight-delta forensics (807 tensors)	non-target drift 0.17%; q/v median 21.6%, max 121%	merge machinery correct; lr 1e-3 runaway is the cause
Checkpoint sweep (r2)	damage present from step 400	no salvageable checkpoint (B5-SWEEP-2026-001)
Recipe probe (gentler lr, shared)	slows, does not cure	shared adapter is structurally interference-bound
yemba per-language probe	2.3 h single-word data vs conversational eval	data-limited; training cannot fix; data-first track
kinyarwanda probe (300 h)	clean U-curve, best 0.375 @ 1800 steps	trainable; scaled decisive run justified
kinyarwanda decisive (300 h, rank 32, 6k steps)	dev plateau 0.355; FROZEN-POOL 0.372/0.089 vs 0.410/0.097	shipped as v1 by owner order at 9.3% rel. gain
kinyarwanda v2 full FT on ~2,000 h	PENDING (data uploading)	ship only if > 9.3% rel. gain

Evidence: platform/evidence/: B5-T6-GATE-2026-001, B5-DIAG-2026-001, B5-SWEEP-2026-001, B5-PROBE-2026-001/-002, B5-KW-DECISIVE-2026-001, B5-CAMPAIGN-R2-TERMINAL-2026-001; SageMaker job medzen-b5-b5-kinyarwanda-full-2026-001 status Completed (listed 2026-08-20).

11. Significant problems, root causes, resolutions

#	Problem	Root cause → resolution
1	Eval attempt 25 failed at container start	asset cards resolve absolute /models paths; pod mounted /input only → /models subPath mount + render-
2	T6 gate box failed 7× (each fail-closed, <$1 each)	exact-key fetches (no ListBucket) · ECR needs SG self-trust · image ENTRYPOINT → --entrypoint python
3	Adapter merges wrong / rank mismatch	wrap_lora targeted llama_decoder by default → CTC_SCOPE_PREFIX='encoder.'; hardcoded rank 16 vs
4	English ingest box died repeatedly	IAM propagation (wait + verify creds INSIDE box), missing KMS decrypt on repo tarball, ProfileNotFound (
5	Full CV kinyarwanda pull refused at scale	identical audio bytes with different transcripts → explicit opt-in MEDZEN_BYTE_CONFLICT_POLICY=dro
6	Bedrock ValidationException on model id	bare Haiku id needs provisioned throughput → eu.anthropic.… inference profile (config only)
7	Gateway image builds failed	service-dir build context vs repo-root COPYs → repo-root context; Dockerfiles enforce SOURCE_COMMIT
8	Fish secret name wrong in code+runbook+ima[illegible]	[illegible] 2026-08-20 by [illegible]'s verification: medzen/fish-api-key does not exist → default now medzen
9	NV watcher died at its own 10 h ceiling	ingest itself healthy; re-armed with 30 h horizon — lesson: watcher timeouts must exceed the watched pro

12. Architecture / implementation decisions Claude changed, and why

• **Shared multilingual adapter → per-language adapters** (B5-TRAINING-STRATEGY-2026-001): T6 + forensics showed cross-language interference is structural, not a tuning artifact. French/swahili/lingala already meet bars untrained (base model serves them); yemba/ewe/pulaar are data-limited → data-first.
• **LoRA-only → LoRA + full fine-tune mode**: owner chose option B (full FT on all ~2,000 h kinyarwanda) after LoRA plateaued at 0.355 dev; v1 shipped now, v2 must earn its promotion (>9.3%).
• **kinyarwanda bar 0.30 → 0.38 abs + 0.09 rel**: owner-revised on decisive-run evidence; verbatim quotes preserved in the gate file.
• **Mock-only gateways → real providers behind env gates** (owner order): fail-closed defaults kept; fake modes remain the default everywhere.
• **Eval-only Tier B data → training-unlocked** on the owner's private signed agreements, recorded as attestation-only decision records (LIC-2026-003/-004) — the yemba_egra precedent.
• **Byte-dup hard refusal → explicit opt-in drop_both** at scale (never train on contested labels; counts logged for review).

13. AWS resource inventory (all eu-central-1, account 558069890522; verified 2026-08-20)

S3 — single data bucket **medzen-speech** (SSE-KMS key 9c336116-c648-4548-95c6-1b926478ae57; any new role needs kms:Decrypt + GenerateDataKey):

Prefix	Contents
approved/kinyarwanda/asr/v1/model.pt	the shipped v1 model (1.95 GB)
curated/<lang>/asr/<config>/v1/	24 language prefixes; per-corpus audio + A3 manifests (e.g. nv_hausa, cv17_rw, av_pcm pending)
eval/<language>/	FROZEN evaluation pools (40+ languages) — never write here; training data must never byte-match the
raw/_incoming/africanvoices/	owner-uploaded Pidgin archives (7/42 landed at report time; box consumes + deletes)
research/b5-training/	job workspaces, eval bundles, box logs (rw-ingest/, av-ingest/, t6-eval-r2/)
curated/_versions/	dataset generation (gbN) version bindings

ECR images (speech platform; digest prefixes):

Repository:tag	Digest	Role
medzen-orchestrator:v1-2026-08-20a	1d970e6f…	B6 orchestrator
medzen-rag-index:v1-2026-08-20a	bc882db3…	RAG index service
medzen-llm-gateway:v1-bedrock-2026-08-20a	cfc677dd…	LLM gateway (Bedrock)
medzen-speech-tts-gateway:v1-fish-2026-08-20e	4dfbf917…	TTS gateway — SUPERSEDES -20d/-20c; adds MEDZEN_FISH_SECRET_ID
medzen-asr-runtime:b6a-003c-89f94d3	47d86776…	ASR runtime
medzen-model-loader:b6a-003b-69bdce1	ac6f694b…	weights initContainer
medzen-asr-eval-runtime:pilot-8f63996	7f9938e0…	eval harness (gate runs pin sha256:dafc4b14…)
medzen-trainer-omniasr:ft-2026-08-19a	fb15770e…	trainer (LoRA + full-FT modes)

Compute / other:

Resource	Id / name	State & purpose
EC2 (running)	i-04ad7c0f8cb06e9db c6i.4xlarge	rw 2,000 h ingest; self-terminates on completion
EC2 (running)	i-03cd4aa5fd4ff1519 c6i.4xlarge + 1 TB gp3	Pidgin staged ingest; self-terminates; 40 h staging deadline
EKS	cluster medzen-speech; ASGs eks-cpu 0/0/4, eks-gpu 0/0/1	control plane running (≈$74/mo), zero nodes — serving is dark
SageMaker	quota L-56AE9D73 = 1 (increase to 2 PENDING, case c060a64[illegible]	[illegible]-b5-kinyarwanda-full-2026-001 Completed
ECS (hands off)	cluster medzen-tts-dev, service medzen-tts-gateway	OWNER'S LIVE reference TTS — review only, never modify
IAM (ephemeral)	medzen-av-ingest-ephemeral (role + instance profile)	tear down after Pidgin ingest completes
SSM	/medzen/registry/tts/voices; /medzen/registry/test/b6/<sha>/*	TTS voice registry (kinyarwanda approved, pidgin eval-only); B6 test registry
Secrets	medzen/client-api-keys (HAS VALUE); medzen/speech/fish-api-[illegible]	[illegible]); medzen/tts/dev/fish-api-key (live dev's own
Bedrock	eu.anthropic.claude-haiku-4-5-20251001-v1:0	LLM model (eu inference profile — bare ids are refused)
Networking	subnet-01fb2fc3f56bce55e; SGs sg-0fee72d218ac002a7 + sg-01[illegible]	[illegible] VPC endpoints, SG self-trust for ECR)

14. Important local directories and files

Path	Contents
~/Documents/medzen-platform/	THE repo (HEAD 6d7d68c). pipeline/ (ingest + 14 adapters + omniasr_train.py), services/ (7 services), scrip
platform/runbooks/B6-DEPLOY-RUNBOOK.md	THE deploy procedure + 2026-08-20 addendum: images, env contract, owner MANUAL ACTION for the Fis
registry/gates/*.yaml	per-language promotion bars; kinyarwanda-v1.yaml carries the owner's verbatim overrides
~/Documents/medzen-shared/codex_status.txt	[illegible] append status here; authorization phrases for training jobs live here
~/Documents/medzen-dropbox/africanvoices/	42 Pidgin archives (154 GB) — the Mac-side source being uploaded; keep until S3 copy verified complete
Session scratchpad (/private/tmp/claude-501/…	[illegible] scripts, watcher logs (av-upload.log, naijavoices-ingest.log) — session-lived, not durable

15. Cost & running-resource status

Cost Explorer (month-to-date Aug 1–20, lags ~24 h): gross recorded usage ≈ $50 (ECR $20, ECS $11, EC2-other $8, RDS $7, S3 $2.5), fully offset by credits — net ≈ $0. The two ingest boxes + SageMaker jobs of the last 48 h are not yet fully reflected.

Currently accruing: 2× c6i.4xlarge (≈$1.36/h combined, on-demand) + 1 TB gp3 (≈$0.11/h) — both boxes self-terminate; worst case if both run 3 more days ≈ $110, reported maximum ≤ **$210** (owner rule: +$100 on every reported max). EKS control plane ≈ $74/mo. ECS medzen-tts-dev and an avatar-chatbot GPU ASG (1/1/1) belong to the owner's other workloads — not this project's engineering, not touched.

Upcoming approved-path spend: kinyarwanda v2 full FT ≈ $70–150 on-demand, reported maximum ≤ **$250** — launches only behind an owner-authorized packet.

16. Known issues, risks, technical debt, assumptions

• **Fish key (RESOLVED 2026-08-20):** investigation of the live ECS medzen-tts-gateway found the key in Secrets Manager medzen/tts/dev/fish-api-key (the service passes the ARN via env FISH_SECRET_ID and fetches at runtime). The platform gateway now takes MEDZEN_FISH_SECRET_ID at deploy to reuse that same key with zero credential handling (grant the pod role GetSecretValue on it); alternatively the owner may still load medzen/speech/fish-api-key for namespace separation. No longer a blocker.
• **Owner decision pending:** serving operating shape (always-on / business-hours / on-demand) — blocks task #37.
• akan/serer baselines known-wrong (#35). LLM calibration unrun (#36).
• Orchestrator pins the TEST registry namespace (/medzen/registry/test/b6/) — production namespace is a deliberate, reviewed deploy-day change; do not quietly widen it.
• Quota increase in human review — plan as if 1 slot until it flips.
• Wave-2's owner-approved 13-language list predates the per-language pivot — re-scope before spending.
• pidgin voice in the TTS registry is EVAL-ONLY (approved:false) — the gateway refuses it for production use by design.
• Watchers/loops in the current Claude session die with the session; the boxes and uploads themselves do NOT depend on the session (only the NV Mac-side queue and the AV upload run on the Mac — keep it awake).
• Assumption baked into ingest: usable() bounds 1–30 s drop very long spontaneous Pidgin takes (metadata shows up to 467 s); drop counts are logged per batch — revisit with segmentation if yield is too low.

17. Exact current task and where work stopped

Work stopped mid-arc "**prep NaijaVoices + Pidgin for the next training session**" (owner-approved plan, in execution): the Pidgin adapter is committed and its ingest box is live in its staging loop; NV igbo is uploading from the Mac; the rw 2,000 h re-pull is uploading from its box; the quota request is pending. **Nothing is blocked on engineering** — the next engineering actions all trigger on completions: (a) rw upload completes → curate v2 → gb5 → v2 packet to owner; (b) Pidgin ingest completes → Tier-B review → decision table; (c) quota flips → note the second slot in the packet.

18. How to verify the current system before changing anything

```
cd ~/Documents/medzen-platform && git log --oneline -15 # recent work
.venv/bin/python -m pytest tests/ -q # expect 2202 passed
export AWS_PROFILE=medzen # all commands below
aws s3 ls s3://medzen-speech/approved/kinyarwanda/asr/v1/ # shipped model
aws s3 cp s3://medzen-speech/research/b5-training/rw-ingest/host.log - | tail -20
aws s3 cp s3://medzen-speech/research/b5-training/av-ingest/host.log - | tail -20
aws ec2 describe-instances --filters Name=instance-state-name,Values=running \
--query 'Reservations[].Instances[].[InstanceId,InstanceType,Tags]'
aws sagemaker list-training-jobs --sort-by CreationTime --sort-order Descending --max-results 5
aws service-quotas list-requested-service-quota-change-history-by-quota \
--service-code sagemaker --quota-code L-56AE9D73 --region eu-central-1
aws ecr describe-images --repository-name medzen-speech-tts-gateway \
--query 'sort_by(imageDetails,&imagePushedAt)[-1].imageTags' # expect -20e
aws secretsmanager describe-secret --secret-id medzen/speech/fish-api-key \
--query 'VersionIdsToStages' # null = still EMPTY (owner must load)
```

```
tail -40 ~/Documents/medzen-shared/codex_status.txt # engineering log
```

19. HANDOFF — START HERE

Inspect first (30 minutes): (1) tail of ~/Documents/medzen-shared/codex_status.txt — the log the owner reads; (2) the two box logs (§18 commands) — they decide your first action; (3) registry/gates/kinyarwanda-v1.yaml — the owner's verbatim decisions you must honour; (4) platform/runbooks/B6-DEPLOY-RUNBOOK.md — the deploy contract; (5) run the test suite.

Do NOT redo: the frozen baselines (suite attempts 29–44), the v1 gate, Tier A ingests + review, the five service images (only rebuild on code change, always with SOURCE_COMMIT), the wave-1 forensics — the conclusions are evidence-backed and owner-ratified. Do not touch ECS medzen-tts-dev. Do not commit anything referencing the owner's private signed agreements beyond the existing attestation records.

Highest-priority task: the kinyarwanda v2 chain. When the rw box logs INGEST_EXIT_0: verify the v2 manifest in curated/kinyarwanda/asr/cv17_rw/v2/, curate (drop_both counts, byte-leak strip vs the frozen pool), bind gb5, then draft the v2 full-FT packet and put the numbers to the owner in the shared file. Launch only after the owner's explicit authorization phrase. Ship as v2 only if relative gain > 9.3% — this is a standing owner order.

Second: when the AV box logs INGEST_EXIT_0, run the Tier-B review (NV hausa/igbo/yoruba + Pidgin), then present the next-wave decision table (bars + costs) to the owner. Tear down the medzen-av-ingest-ephemeral role/profile afterwards.

Owner-only items to surface, not perform: decide the serving operating shape; approve training numbers. (The Fish key needs no owner action anymore: deploy with MEDZEN_FISH_SECRET_ID=medzen/tts/dev/fish-api-key, or the owner may optionally copy it into the platform namespace.)

Working discipline that kept this project safe: verify from inside the box before trusting it; fail closed on anything ambiguous; one authorization = one numbered attempt; every gate result becomes a committed evidence record; report costs on-demand +$100 on maxima; and when a claim matters, re-verify it live — that is how §11 item 8 was caught the day before it would have broken the deploy.